CSE408 Software Engineering
Software Design Phase

Group - 16
OptiLearn

Group Members

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1. Introduction

1.1 Purpose

OptiLearn is an online productivity application that can be used by students to improve their study habits. This documentation will be a guide to the devs, stakeholders and evaluators to ensure they have a glimpse of all the features, behavior as well as the constraints and that the app is properly constructed.

1.2 Scope

OptiLearn allows user to plan day-to-day study, record what we are actually doing and obtain predictive insights based on the data to modify. You can make plans, choose subjects and topics and approximate time and later take notes about your actual session, concentration and other distraction. The system contrasts the planned and actual ones to define the effectiveness, identify productive periods, and indicate exhaustion or burnout. On the basis of such an analysis it gives individualized recommendations clinical optimum study times, session lengths, the order of subjects, etc. With time it gets to know your habits to assist in reducing procrastination, being consistent and again to study better.

2. System Overview

The system has major functional units which are as follows:

- Registering and Authentication of users - Encrypted sign-up and sign-in.
- Creation of Study Plan - Add the subjects and the topics, assess the study difficulty and plan the durations of study.
- Study Session Tracking -Record real study sessions including starting/stopping time or duration, subject studied, self-reported focus and freeform notes.
- Productivity & Efficiency Analysis– Compare the planned vs actual time in studying to identify the high efficiency and low efficiency slots.
- Smart Recommendations – Work with an LLCM API to create personalized insights, such as recommendations on the optimal time slot to do anything, chunking work tasks, and determining the optimal session duration.

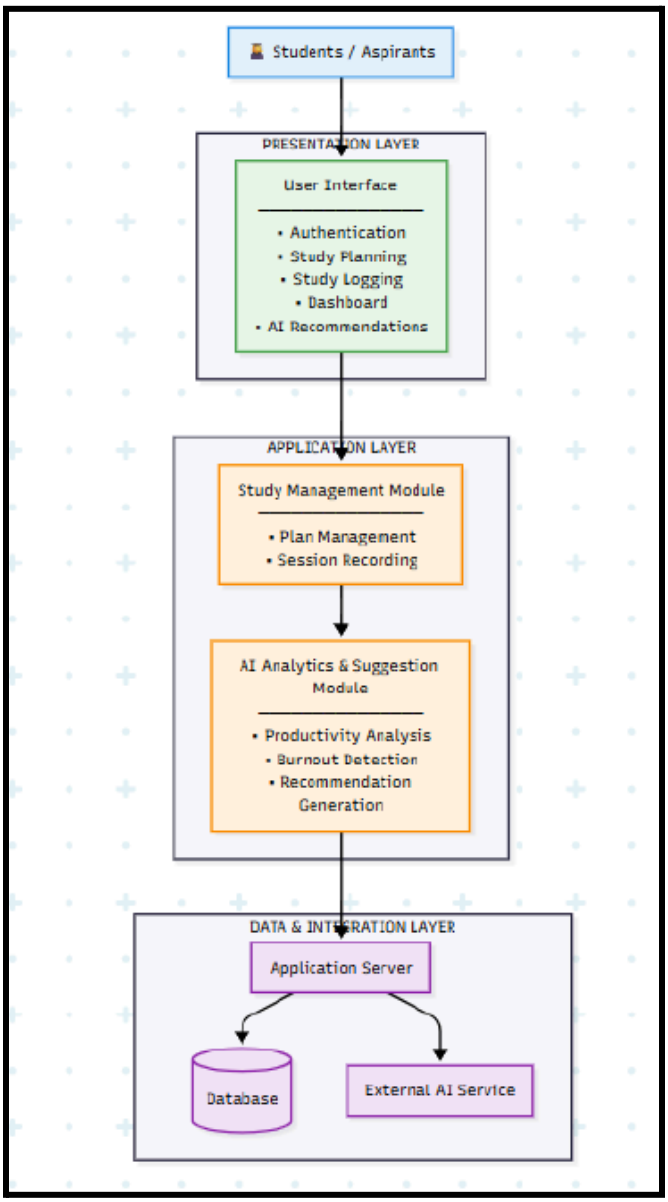
3. System Architecture Design

3.1 Architecture Overview

- **UI Layer -** The UI Layer is a simple interface that I use on a daily basis. It manages getting the log-in process on, arranging my study program, recording when I learned, viewing my dashboard, and presenting me with AI suggestions. It simply transports what I type into it and provides feedback on it and it does not store data anywhere.
- **Authentication Layer -** User account is secure due to the Authentication Layer. It verifies user's details and maintains my session to ensure that no other person can access my study data or analytics. It is a fundamental safeguard of user information, in fact, it keeps everything confidential.
- **Study Management Layer -** The Study Management Layer is the place where user record his/her study plans and study sessions. It defines what user was going to study, the duration he/she studied and the subjects and topics covered. It systematizes all these data, so as to enable and review what user progress is.

- **Analytics & Suggestion Layer -** The Analytics and Suggestion Layer review user's study history and transforms it into actual productivity. It monitors user efficiency, identifies trends in work, reminds user about the risk of burnouts, and provides personalized suggestions. It puts raw data on real-world advice that may help in remaining steadfast and perform better.
- **Data Layer -** All user's data is found in the Data Layer. It stores user profile, study notes, reports and previous recommendations. It is also the one that coordinate with AI to create personalized insights by ensuring that data is left intact, trustworthy, and shared securely across the subsections of the system.

3.2 System Architecture Diagram



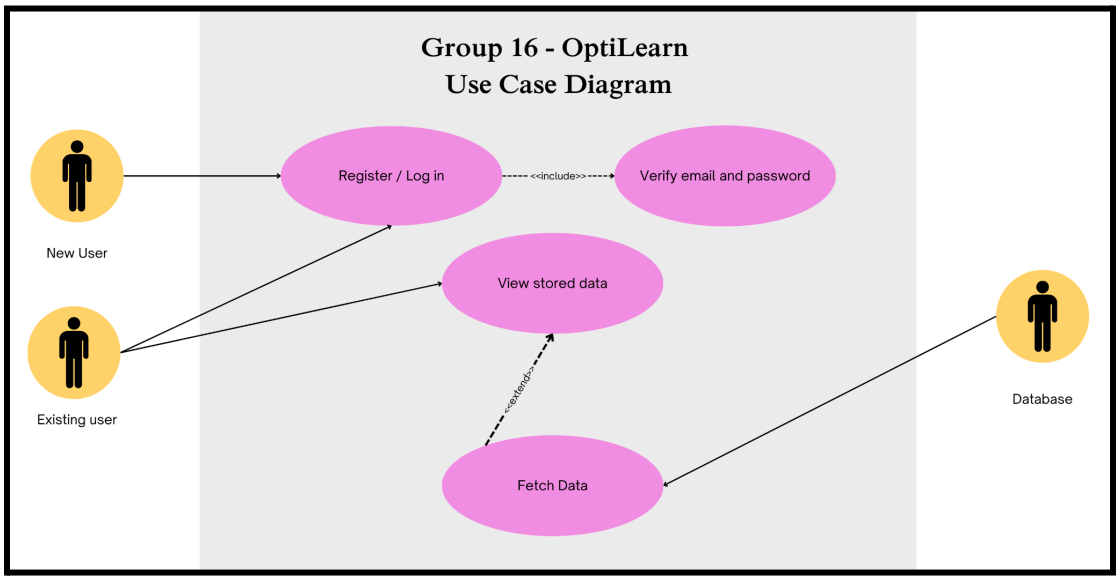
4. Use Case Diagrams

4.1 Overview

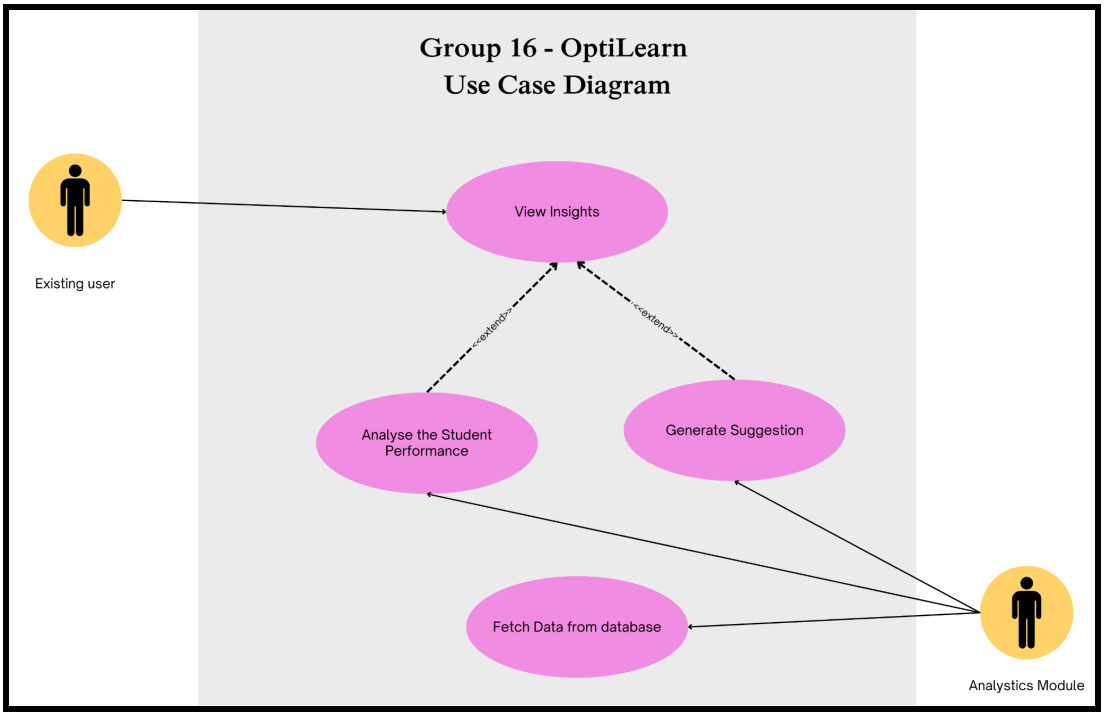
- The OptiLearn Use Case Diagram is more or less a fast overview of the interaction between various users and the essence of the system. It demonstrates the connectedness between the key participants- Students and Aspirants and the System Boundary, which identifies the key functions the software must execute.

4.2 Use Case Diagrams

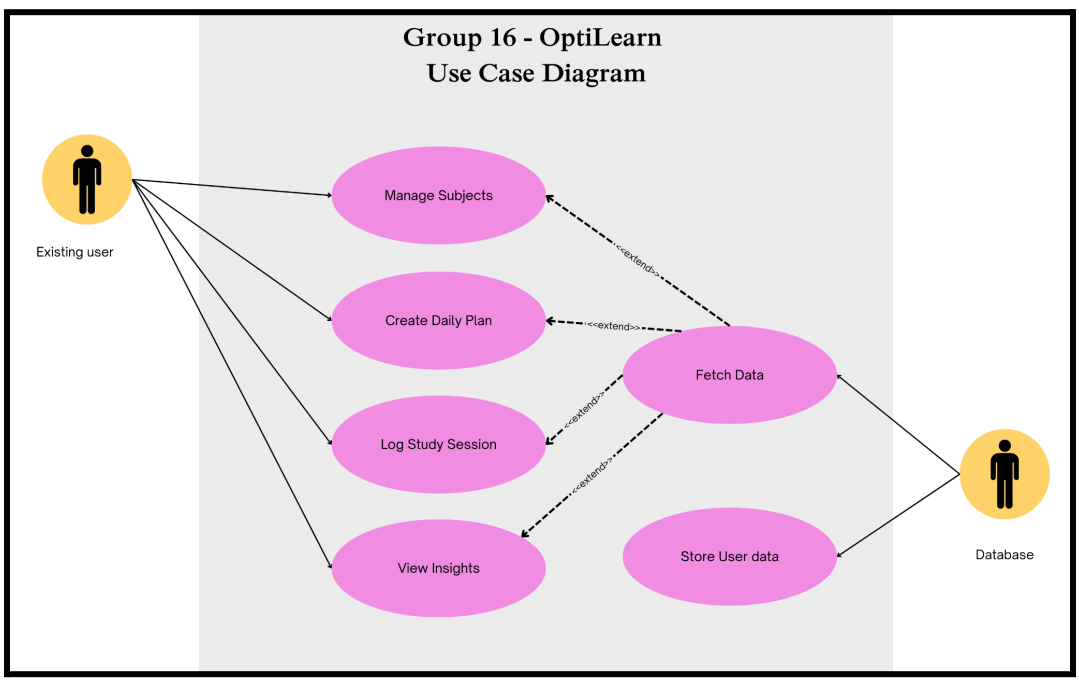
- The initial access control is also configured using this diagram to show how interested users log in and register either as new or existing users. It deals with the inner authentication procedures that are needed to perform authentication and demonstrates how authorized users can access and read their own stored information in the database.



- The use case is centered on the insights functionality on the platform, which illustrates what occurs on the background as a user requests to see their performance analytics. It emphasizes the work of an Analytics Module actor, who retrieves database records to process the performance of students and provide individual study recommendations.



- Lastly, this diagram describes the key interactive capabilities of a current user, including: study subject management, creation of daily plans, recording sessions and insights. It traces these front-end user actions onto the required backend data operations illustrating how the majority of them lead to retrieving or storing data in the base data depository.



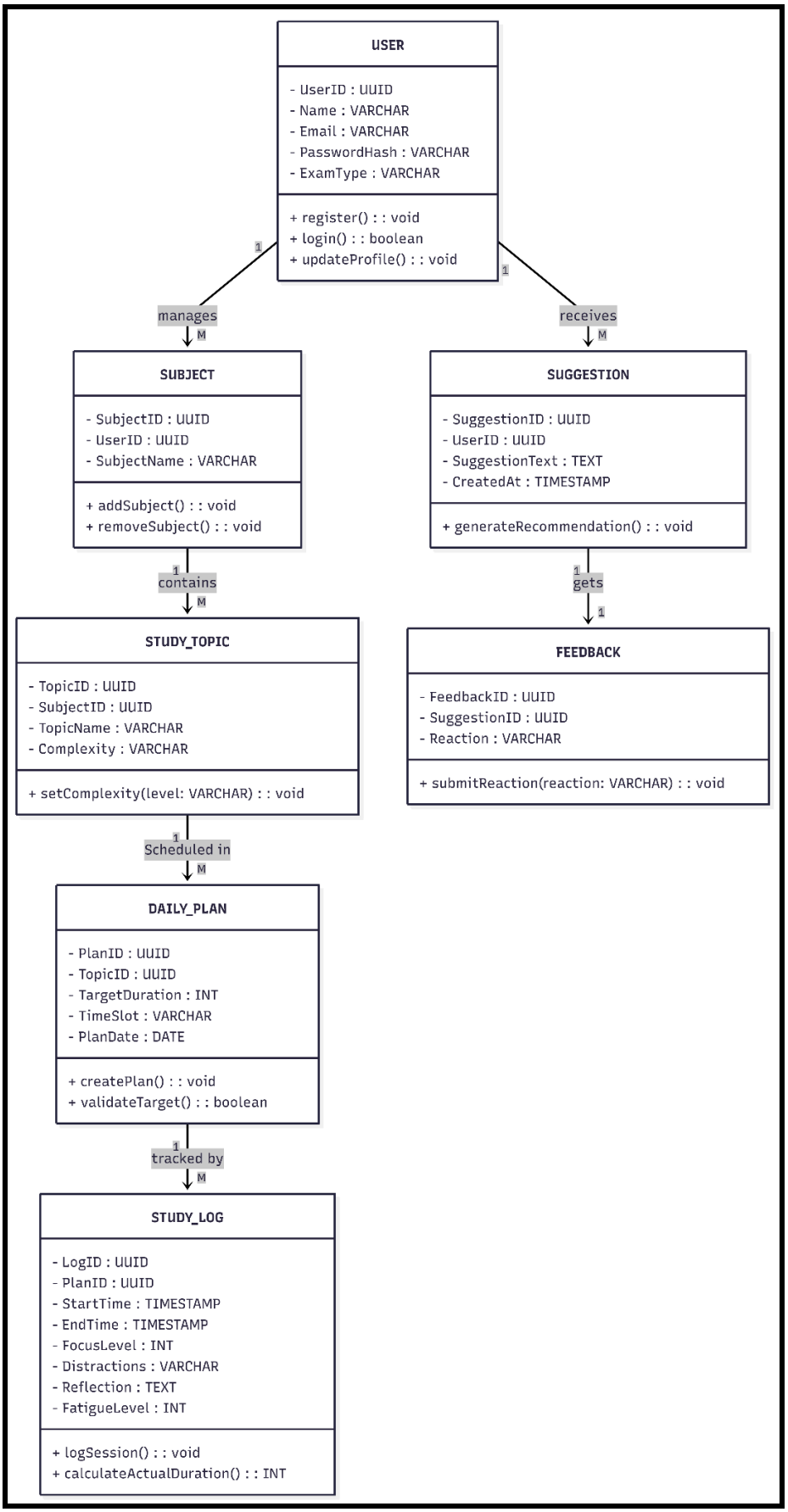
5. Class Diagram

5.1 Overview

- The class diagram of OptiLearn simply represents the entire framework of the system. It enumerates all the classes, their attributes, methods and the relationship between them. Just imagine it as a blueprint, which a developer can consult to understand how the data objects, such as User, StudySession and Topic, communicate in the software.

5.2 Class Diagram

- This UML class diagram will present the key component namely the USER that manages the academic burden through the formation of a SUBJECT. The same is divided into smaller STUDYTOPICS. The subjects are then arranged in a DAILYPLAN and kept track of in a study log- STUDY-LOG. At the same time, the system comes up with a SUGGESTION of what the user can reply with FEEDBACK. The diagram has been enriched with modes like +logSession(), +generateRecommendation that illustrate what each of the classes is able to do.


USER
- UserID : UUID
- Name : VARCHAR
- Email : VARCHAR
- PasswordHash : VARCHAR
- ExamType : VARCHAR
+ register() : : void
+ login() : : boolean
+ updateProfile() : : void
1
manages
M
1
receives
M
SUBJECT
- SubjectID : UUID
- UserID : UUID
- SubjectName : VARCHAR
+ addSubject() : : void
+ removeSubject() : : void
SUGGESTION
- SuggestionID : UUID
- UserID : UUID
- SuggestionText : TEXT
- CreatedAt : TIMESTAMP
+ generateRecommendation() : : void
1
contains
M
1
gets
1
STUDY_TOPIC
- TopicID : UUID
- SubjectID : UUID
- TopicName : VARCHAR
- Complexity : VARCHAR
+ setComplexity(level: VARCHAR) : : void
FEEDBACK
- FeedbackID : UUID
- SuggestionID : UUID
- Reaction : VARCHAR
+ submitReaction(reaction: VARCHAR) : : void
1
Scheduled in
M
DAILY_PLAN
- PlanID : UUID
- TopicID : UUID
- TargetDuration : INT
- TimeSlot : VARCHAR
- PlanDate : DATE
+ createPlan() : : void
+ validateTarget() : : boolean
1
tracked by
M
STUDY_LOG
- LogID : UUID
- PlanID : UUID
- StartTime : TIMESTAMP
- EndTime : TIMESTAMP
- FocusLevel : INT
- Distractions : VARCHAR
- Reflection : TEXT
- FatigueLevel : INT
+ logSession() : : void
+ calculateActualDuration() : : INT

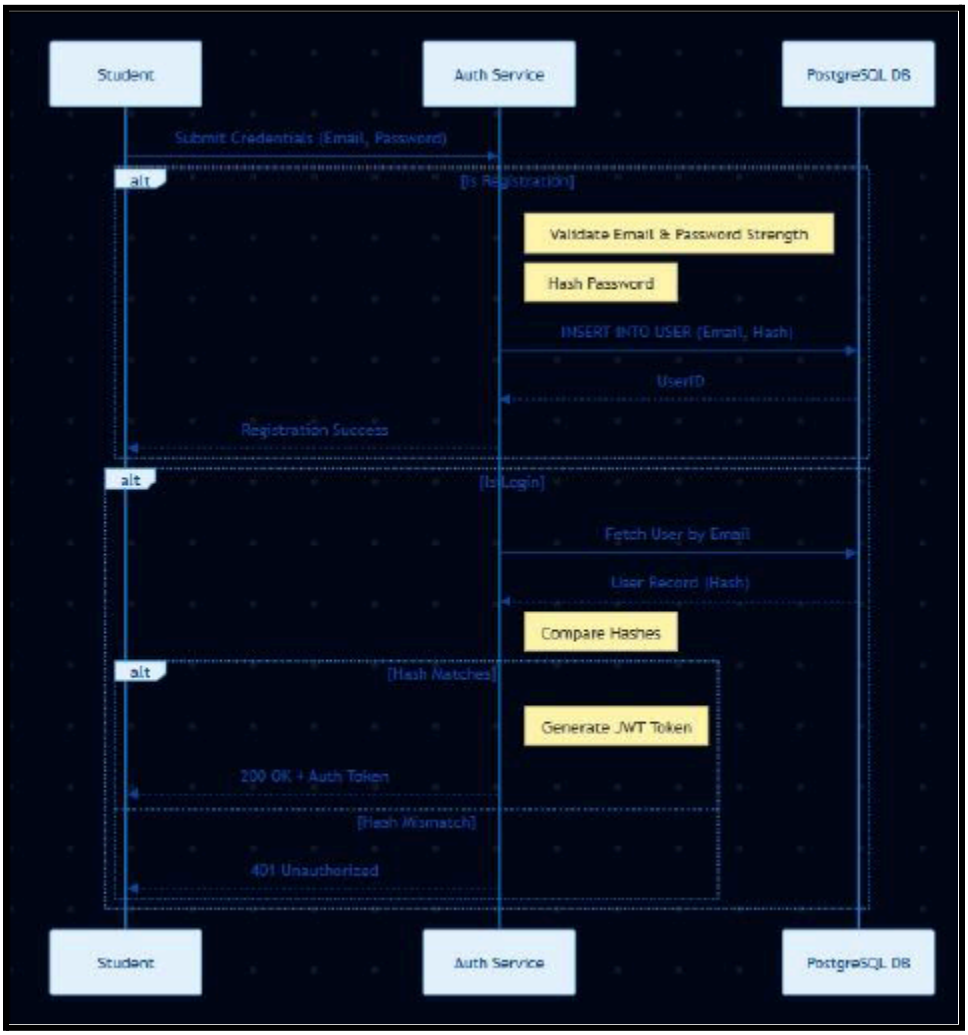
6. Sequence Diagrams

6.1 Overview

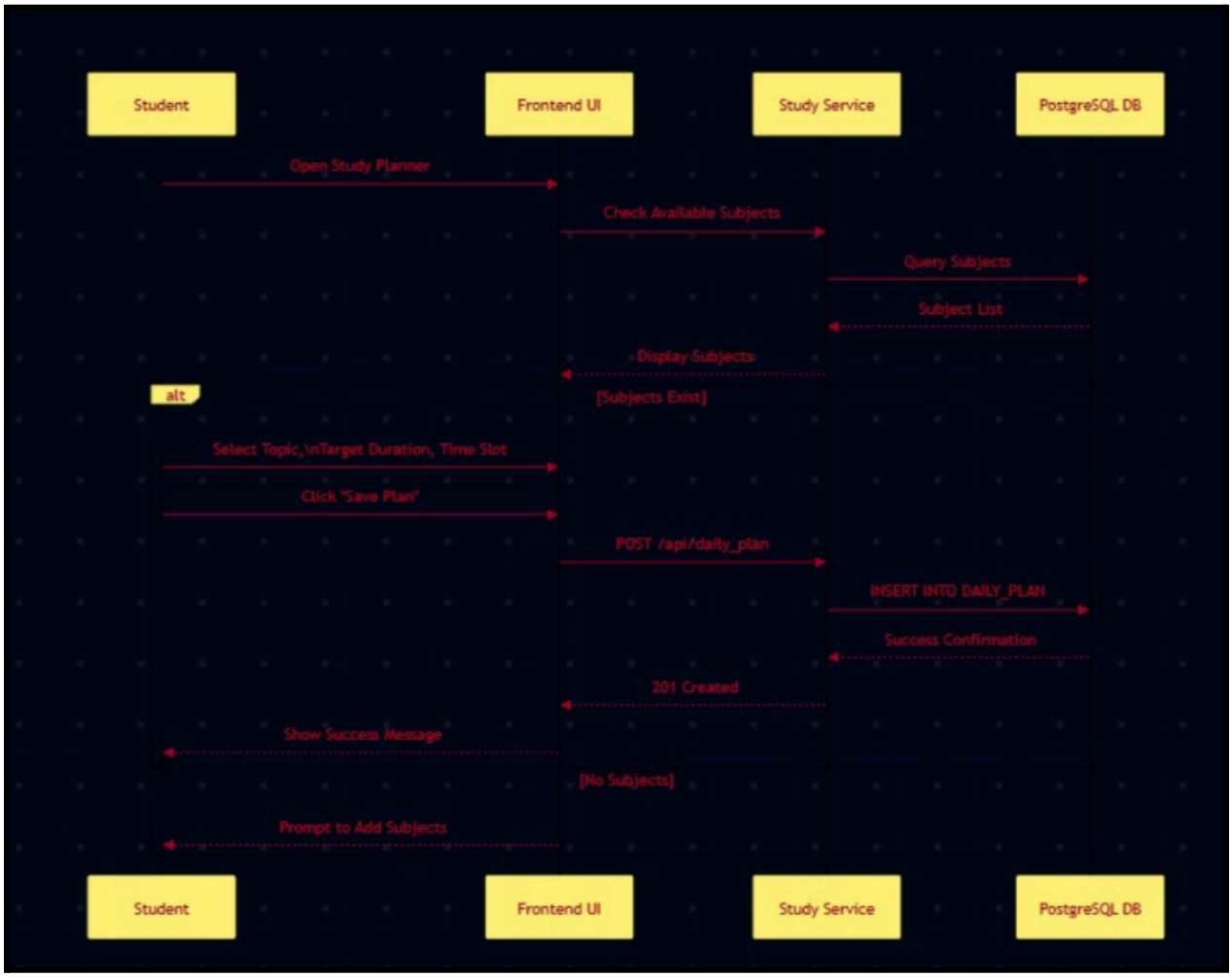
- OptiLearn sequence diagram is used to plan live sequence of interaction of the components of the system within a given duration of time. It magnifies the message between the user and the system modules to and fro until one completes a specific task such as recording a study session.

6.2 Sequence diagram

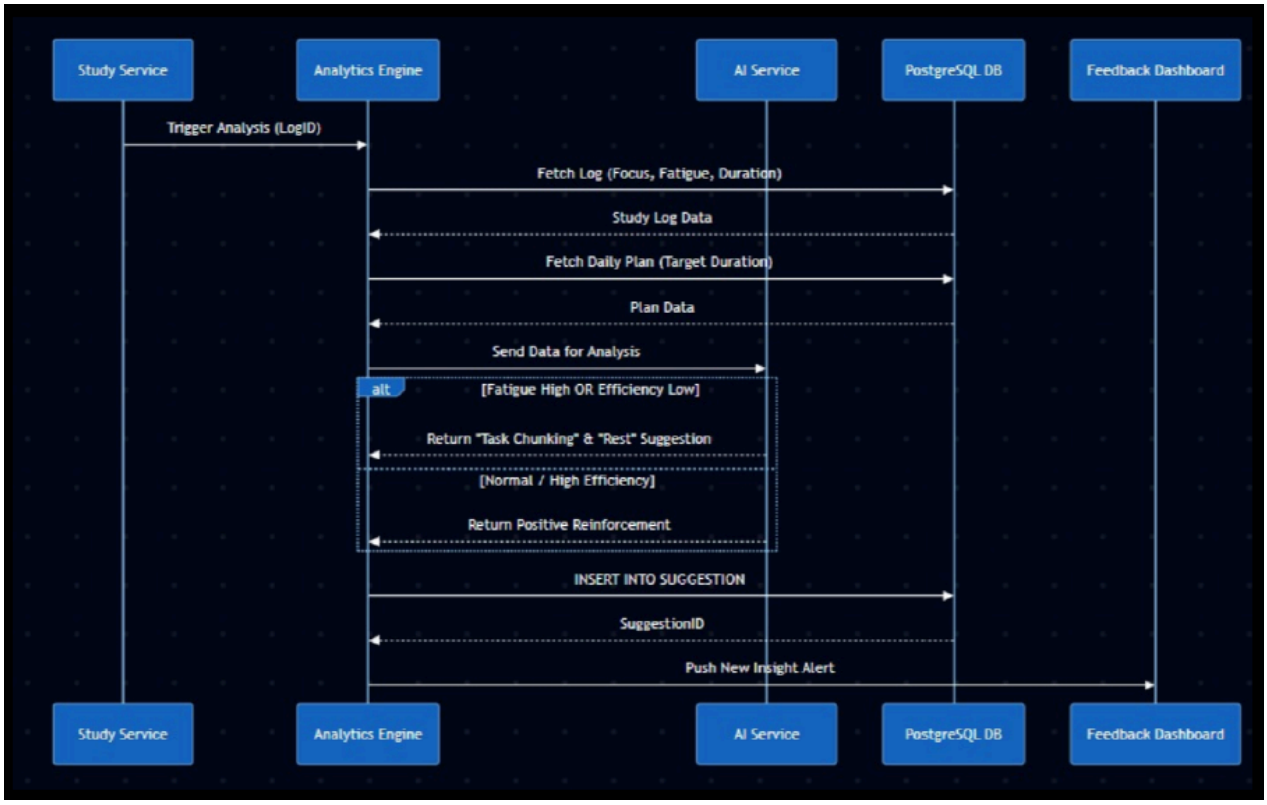
- This sequence diagram illustrations the manner in which OptiLearn functions internally: Authening (hash passwords and spin off tokens), planning studying (course choice and duration of study), and AI analysis. The platform scrapes your performance metrics out of a PostgreSQL DB to simulate a personalized, human-like tip, that is, it is like Task Chunking style advice but with regard to how weary or effective you appear.



- In this sequence diagram, you will visualize the flow when a student organizes his daily study process and in the second one, the AI will be involved as soon as you report your session. The system retrieves your data out of Postgre SQL, scrubs your fatigue and efficiency, and cooks up personalized recommendations, such as Task Chunking, and plops them down onto your dash.



- The final sequence drawing is a mapping of the AI feedback loop of OptiLearn. Upon signing a study session, the Analytics Engine retrieves performance information in the PostgreSQL DB and sends it to the AI Service. Relying on your efficiency score, it either spits out some corrective hints or kudos, saves the note and notifies the Feedback Dashboard.



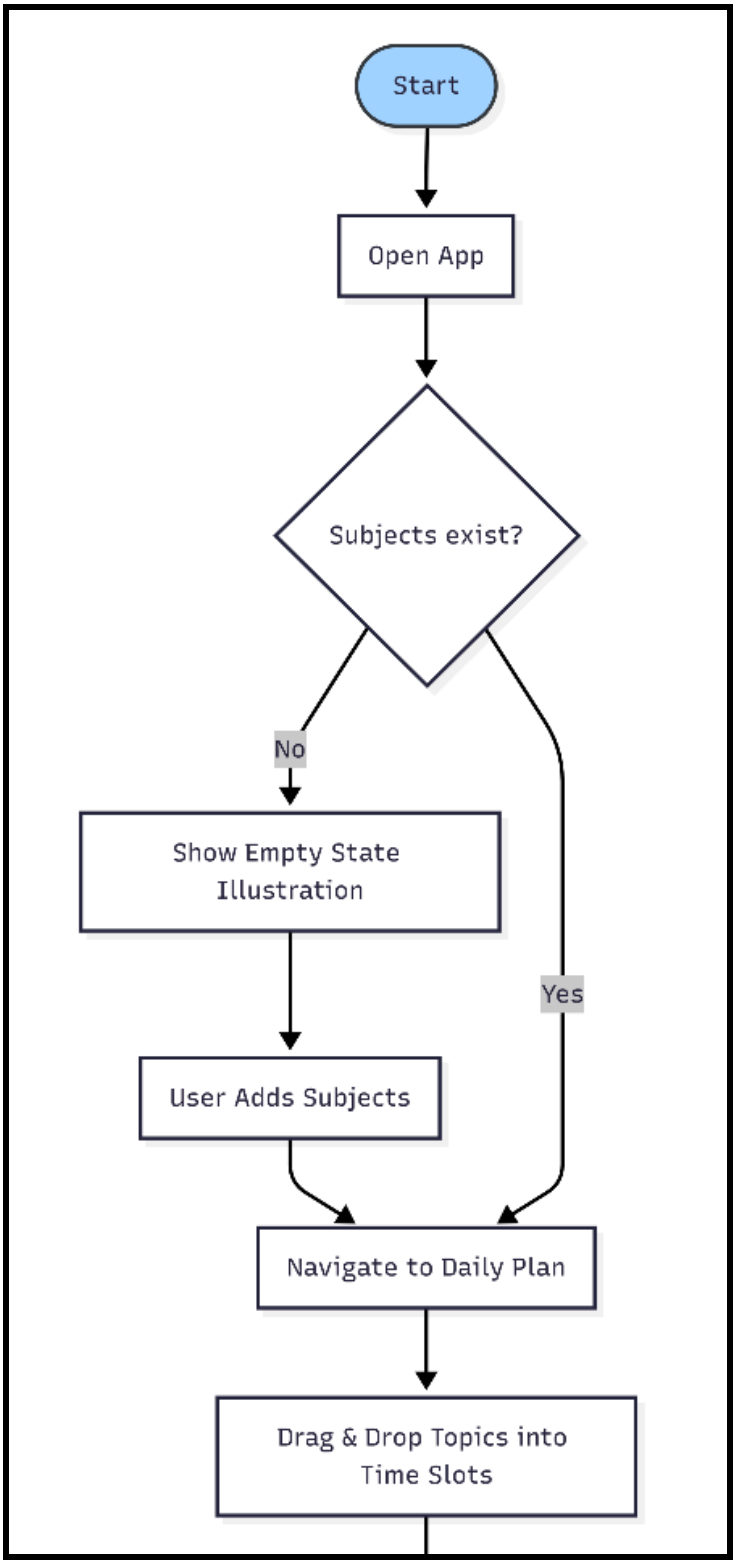
7. Activity Diagrams

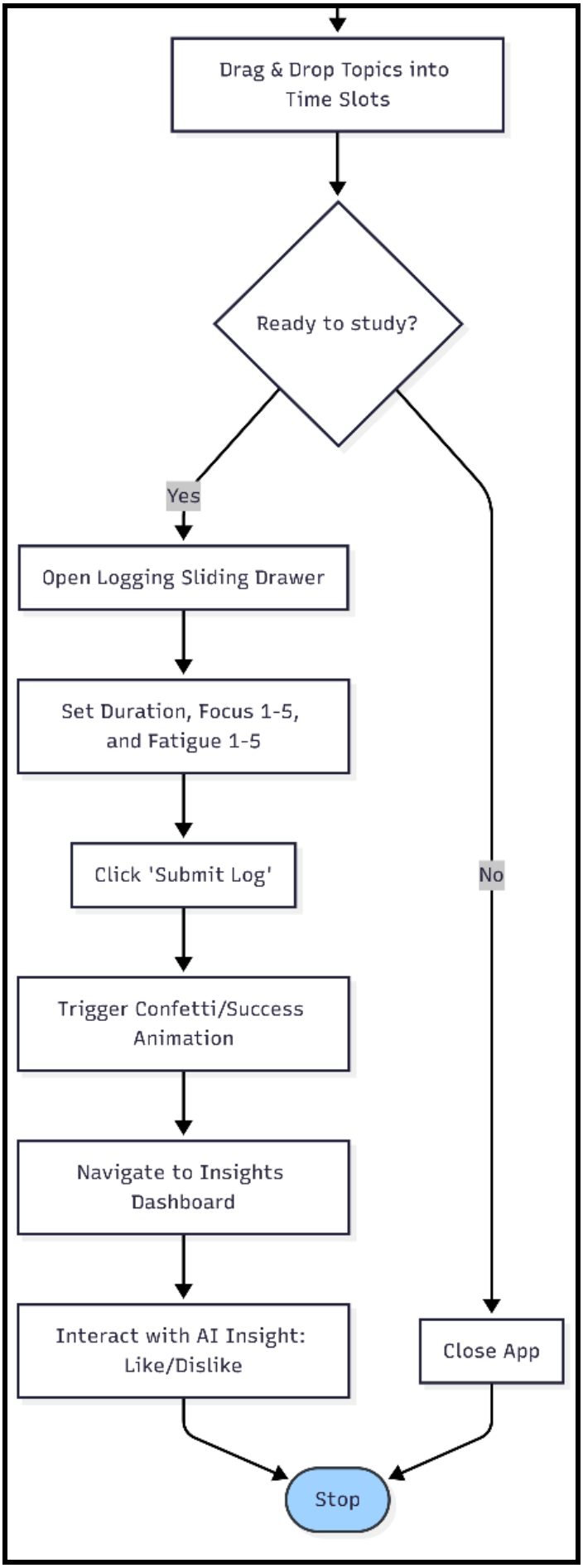
7.1 Overview

- The OptiLearn Activity Diagram essentially is a breakdown or roadmap of how all this works, starting with you clicking on a study thus all the way to the point where it spews out AI knowledge. It is a rather advanced flow chart that works with decision-making and parallel tasks.

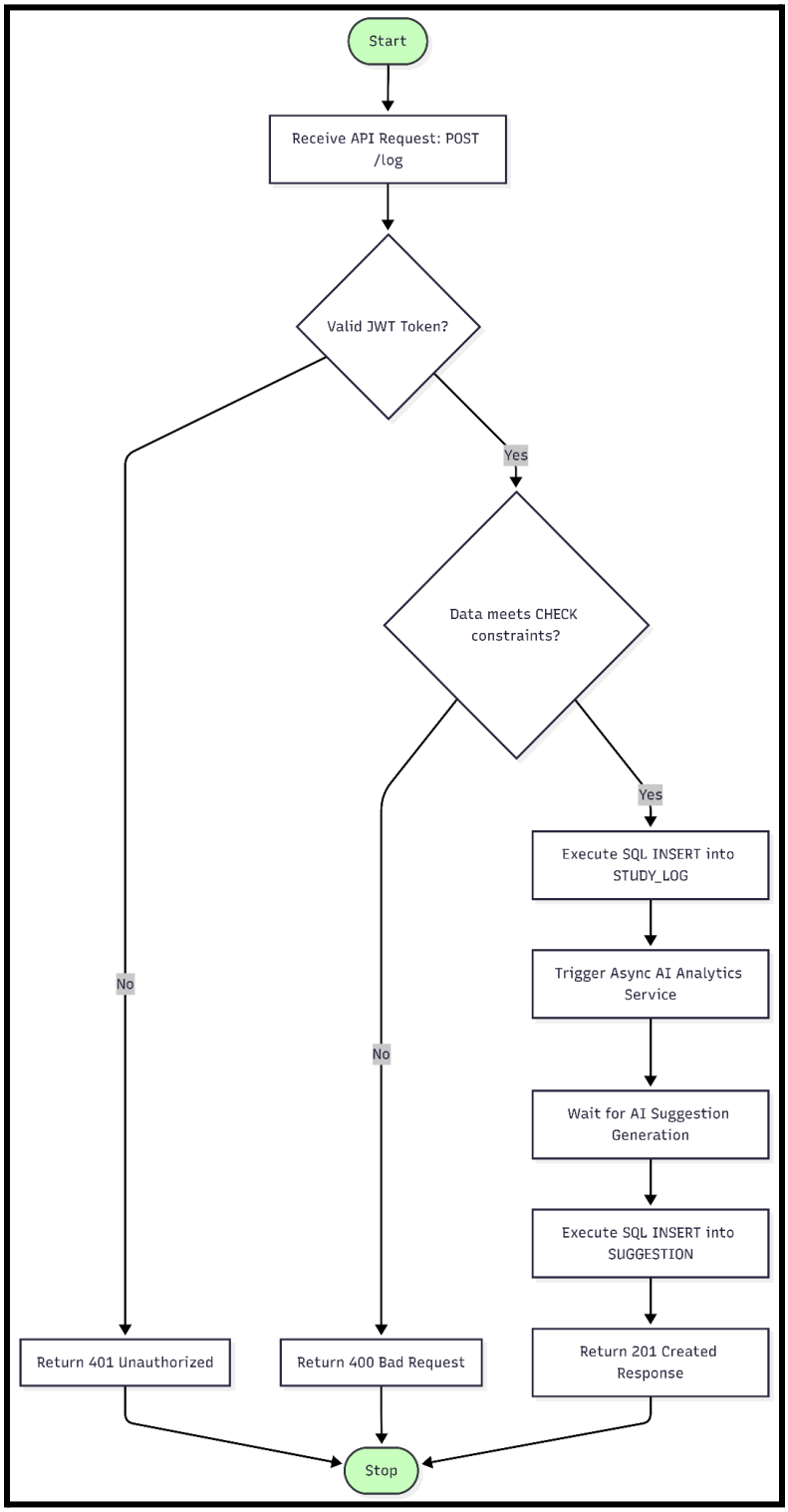
7.2 Activity Diagram

- This flowchart is in the footprint of the front-end: open the application, select your topics and design a study plan by dragging it on a daily basis in your favorite way. It monitors your concentration during a session, your exhaustion, has a stylish success animation, and proceeds to display AI-generated insights to your dashboard.

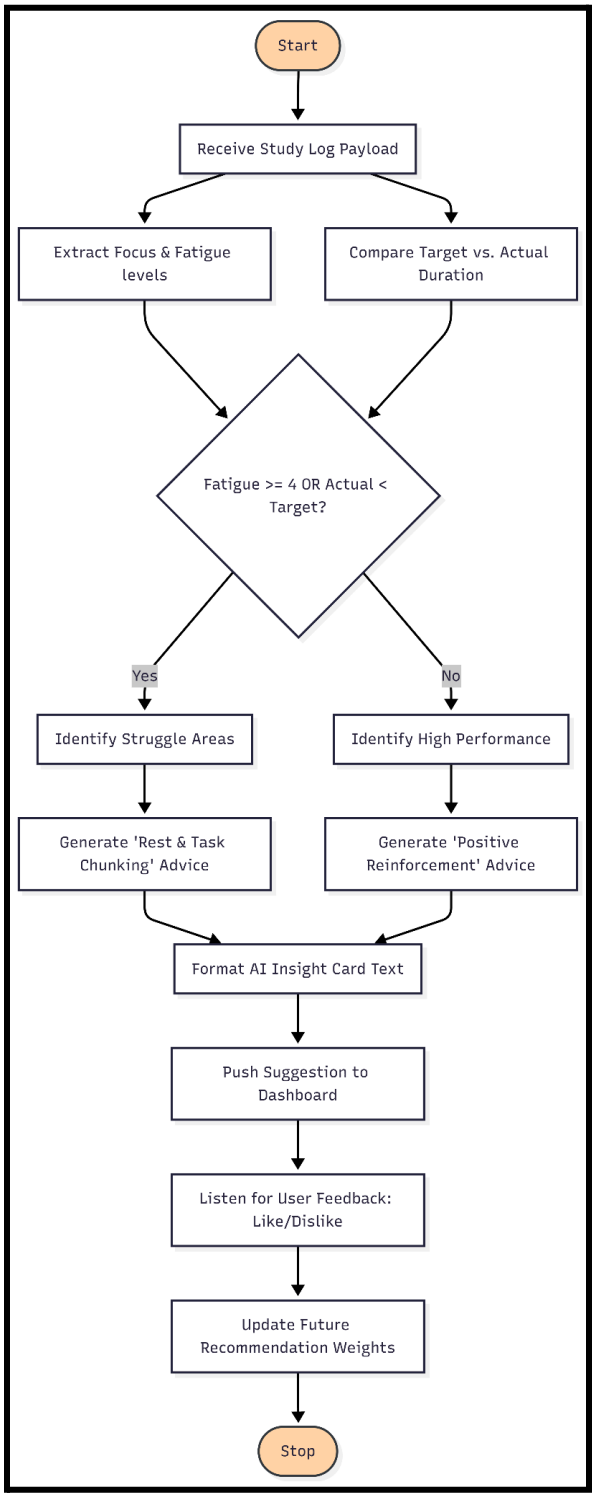

Start
Open App
Subjects exist?
No
Yes
Show Empty State Illustration
User Adds Subjects
Navigate to Daily Plan
Drag & Drop Topics into Time Slots


Drag & Drop Topics into Time Slots
Ready to study?
Yes
No
Open Logging Sliding Drawer
Set Duration, Focus 1-5, and Fatigue 1-5
Click 'Submit Log'
Trigger Confetti/Success Animation
Navigate to Insights Dashboard
Interact with AI Insight: Like/Dislike
Close App
Stop

- The logic on the backend is shown in this flowchart. It begins with authentication of the JWT token and verification of the payload policy followed by log storage to the SQL database, which in turn invokes an asynchronous AI analytics service and lastly, a corresponding set of HTTP status codes is communicated.

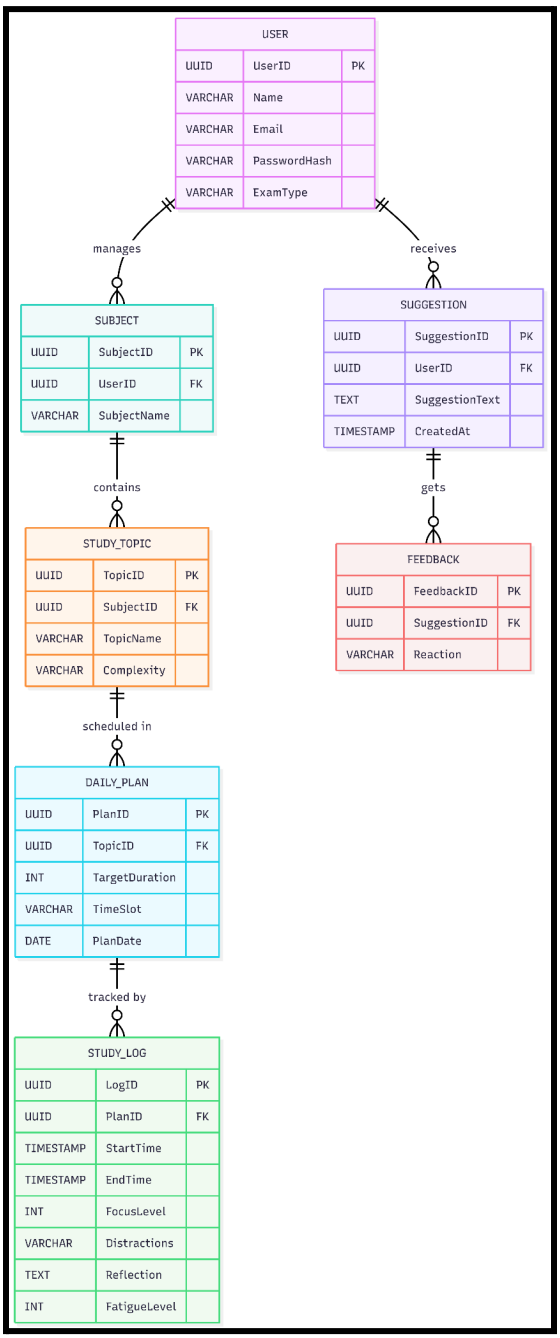


- Finally, the flowchart disaggregates the concept of providing analytical reasoning on personalized study advice depending on the data on focus, fatigue, and duration. It runs off to propose Rest & Task Chunking of high fatigue low duration or Positive Reinforcement when high performance occurs, and the process is completed by a feedback loop to adjust the individual weights in future.



8. Database Design (ER Diagram)

- This diagram is then simply what makes our little theory a real Entity-Relationship Diagram. It defines the apparent one-to-many relationships among the tables (such as one user can have a number of subjects, or one day plan can have many to study logs). It even writes out PKs, FKs and literal types of data such as UUID, VARCHAR and TIMESTAMP such that we never have to deal with data integrity problems.



9. Conclusion

OptiLearn addresses this huge dilemma that most of us encounter when we want to organize our schedules and when we actually intend to accomplish anything. Rather than a dull to-do list, it provides you with an AI-powered, data-driven guidance, which recognizes your high-efficiency time blocks and alerts burnout real-time to make changes on your plan in the fly. Because it is created using a Tech stack of React, Spring Boot, and PostgreSQL, it will remain scalable and secure and allow you to have your academic objectives in check. In brief, OptiLearn provides us the means of reversing these studying behaviors, shedding the daily routine, and striking consistency with smart data analysis as opposed to strict schedules.